

Optimizing Air Ambulance Location and Relocation

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Abstract

Air ambulance services play a crucial role in providing timely medical care, especially in large and geographically dispersed regions. This research aims to optimize the placement of air ambulances by maximizing total demand coverage. To achieve this, we use a multi-period relocation approach. We introduce a mathematical optimization model that allows aircraft to relocate between bases at the beginning of each time period while considering relocation costs. In this study, we consider two types of vehicles and two types of missions that need to be carried out in the system, which aligns with the Ornge air ambulance system, the largest air ambulance service in Ontario, whom we aim to assist.

Keywords: *Air ambulance optimization, Aircraft relocation strategy, Ornge air ambulance, multi-period relocation*

1. Introduction

Air ambulances are a crucial component of emergency medical services, providing rapid transportation for critically ill or injured patients. They are particularly vital in remote and rural areas where ground ambulances cannot reach patients in a timely manner. Additionally, air ambulances play a significant role in urban settings when immediate transportation is required, and ground ambulances are delayed due to traffic congestion.

This study aims to assist in the optimization of air ambulance location and relocation for Ornge, the largest air ambulance service in Ontario, covering approximately 1.08 million square kilometers. As a nonprofit organization primarily funded by the Ontario Ministry of Health, Ornge is responsible for transporting critically ill or injured patients across the province. Ornge handles two types of calls: scheduled calls, which are planned in advance, and unscheduled calls, which occur unexpectedly and require an immediate response. Air ambulance missions are further categorized into two groups: on-scene missions and interfacility missions. On-scene missions typically involve responding to accidents or medical emergencies in remote locations, where only rotor-wing (RW) aircraft are used due to their ability to land in difficult terrain. Interfacility missions involve transporting patients from one healthcare facility to another, such as transferring patients to trauma centers or specialized hospitals. For these missions, both rotor-wing (RW) and fixed-wing (FW) aircraft can be utilized.

In this study, we introduce a mathematical optimization model that uses a multi-period relocation strategy to maximize total expected demand coverage. To balance efficiency and cost-effectiveness, we incorporate a relocation cost penalty to prevent excessive relocations, which can be operationally expensive. Our proposed model aims to maximize the expected coverage of air ambulances by considering two types of vehicles and two types of missions while penalizing relocation distances to prevent excessive relocations. This model will be used to develop another model specifically for Ornge air ambulance location optimization. In this

research, we use a multi-period relocation model, which divides the day into several time periods and optimizes the location and relocation of aircraft at the beginning of each time period to achieve the mentioned objectives.

2. Previous Work

A significant amount of research has been conducted in this area. In this literature review, we examine some early works, deterministic single and multiple coverage models, stochastic and robust location models, multi-period relocation models, online and offline relocation models, fairness in accessibility, and studies specifically related to air ambulance services and Ornge.

One of the first studies on the location of emergency service facilities was conducted by Toregas et al. in 1971. They introduced a mathematical optimization model using a set-covering approach. The model minimizes the number of vehicles while ensuring that each location is covered by at least one vehicle within a predefined distance [1]. Later, in 1974, the Maximal Covering Location Problem (MCLP) model was introduced by Church and ReVelle. This mathematical model aims to maximize the demand covered by at least one vehicle, assuming a fixed number of vehicles. Emergency services sometimes use multiple types of vehicles, such as two kinds of aircraft (rotary-wing and fixed-wing) [2]. The two models mentioned above did not consider this factor, but in 1979, Schilling et al. introduced a model that extends MCLP. Instead of considering just one type of vehicle, their model maximizes demand coverage within a predefined time by both types of vehicles, A and B [3]. Previous models did not incorporate backup coverage, meaning only one vehicle was sufficient to mark a demand as covered. In 1986, Hogan and ReVelle introduced two models, BACOP1 and BACOP2, which incorporated backup coverage into MCLP. The objective of the first model is to maximize the demand covered by at least two vehicles. In the second model, the objective function consists of two parts: maximizing the demand covered by one vehicle and maximizing the demand covered by two vehicles. The first part is weighted by w , and the second part by $1 - w$, ensuring a balance between the two objectives [4]. Later, in 1997, Gendreau et al. introduced the Double Standard Model (DSM), which aims to maximize the total demand covered at least twice within the stricter response time r_1 , subject to the constraints that all demand points must be covered at

least once within the broader response time r_2 ($r_1 < r_2$) and that at least α percent of the total demand is covered within r_1 [5]. In 2015, Su et al. introduced a refined DSM model that incorporates both delay costs and operational costs. Traditional models often use a binary measure, where a demand point is either fully covered or not covered at all. In contrast, this model employs a continuous scale based on the time required for an ambulance to reach a demand point, translating time delays directly into an expected cost [6]. One of the most significant models incorporating probabilistic availability into the Maximal Covering Location Problem (MCLP) is the Maximum Expected Covering Location Problem (MEXCLP). Its objective is to maximize the expected demand coverage while accounting for the probability of vehicle availability. MEXCLP assumes that the probability of an ambulance being busy is p , which is the same for all ambulances in the system. This probability is estimated by dividing the total estimated duration of missions by the total number of available ambulances. For a demand point i covered by m ambulances, the expected covered demand is given by $d_i(1 - p^m)$. The marginal contribution of the m th ambulance is $(1 - p)^{m-1}p$, meaning that the first ambulance

has the greatest impact, the second has a smaller but still important effect, and additional ambulances continue to reduce the risk of non-coverage, though with diminishing returns. The objective function maximizes the sum of these marginal contributions across all demand points to optimize emergency response efficiency [7, 8].

In 2004, Beraldi et al. introduced a stochastic approach to emergency medical service planning. They proposed a stochastic programming model with probabilistic constraints. The model aims to determine the optimal locations for service sites while simultaneously deciding how many vehicles to assign to each site, minimizing both site opening and operational costs. Instead of guaranteeing that all service requests are always met, the stochastic constraints ensure that service requests are met with a prescribed probability level p [9]. Then, in 2009, they introduced a model that uses a two-stage framework with incorporated probabilistic constraints to handle the uncertainty of emergency calls. The model has two parts: the first part, called the first stage, specifies the locations for vehicles by minimizing the costs, and the second stage assigns the calls to sites by minimizing the distances [10].

Another type of relocation model is real-time relocation models. Gendreau et al. introduced a model for real-time ambulance relocation. Similar to the DSM, this model aims to maximize the total demand covered by at least two vehicles while allowing ambulances to relocate after a dispatch. The objective function includes a penalty term that prevents frequent back-and-forth movements of the same ambulance and discourages long trips. This penalty function accounts for the system's relocation history to enforce these constraints [11]. Since real-time or online relocation models need to be solved frequently, after every occurrence, they require significant computational effort. To address this concern, offline relocation models have been introduced. These models optimize ambulance locations for all possible system states in advance. The Maximal Expected Coverage Relocation Problem (MECRP) is a dynamic relocation model introduced in 2006, designed to precompute ambulance relocation plans based on different possible system states. The goal is to maximize the expected demand coverage by summing over all states [12].

Fairness is another important concept in medical services. Most of the reviewed works focus on maximizing total demand coverage, which tends to prioritize densely populated areas, potentially neglecting low-demand regions. Jagtenberg et al. conducted research on fairness in 2020 and 2021 [13, 14]. The second study applied spatial epidemiological methods and mathematical modeling to assess various base location scenarios, considering factors such as population distribution, historical incident data, and geographical barriers. This study compared three social welfare function (SWF) approaches to determine the optimal locations for air ambulances. The Utilitarian SWF, which focuses on efficiency, prioritizes covering larger cities, leaving the northernmost part of Norway largely uncovered. As a result, 13 out of 428 municipalities had a zero probability of being reached on time. The Bernoulli-Nash SWF, which emphasizes fairness, ensures that all municipalities have a non-zero probability of being reached within the response time threshold, embodying a "no man left behind" principle. The Iso-elastic SWF balances efficiency and fairness by introducing a parameter a to control the level of aversion to inequality, creating a spectrum between the utilitarian and Bernoulli-Nash approaches. As a increases, indicating a greater concern for fairness, more weight is given to covering the northern regions of Norway. In 2020, Jagtenberg and Mason introduced a time-sharing approach across

multiple near-optimal ambulance deployment configurations. The proposed model aimed to improve fairness in ambulance planning by distributing time among several near-optimal configurations. It determines the fraction of time to spend in each configuration to maximize Bernoulli–Nash social welfare. This involves a non-linear optimization model that considers the product of individual utilities, representing the probability of timely ambulance arrival, across the population [15]. In 2023, Gunnarsson et al. optimized the location of helicopter ambulance bases in Iceland to find efficient and fair solutions. The study explored the optimal and equitable placement of helicopter ambulance bases by applying both the Maximal Covering Location Problem (MCLP) to maximize coverage and the Fringe Sensitive Location Problem (FSLP) to incorporate fairness. The objective function aimed to minimize the total distance for uncovered demand, weighted by w , while also reducing the total demand not covered, thereby integrating efficiency and fairness simultaneously [16].

Specifically regarding air ambulances, numerous studies have been conducted worldwide. In 2014, Cho et al. introduced a mathematical model to determine the optimal locations for trauma centers and helicopters to maximize the effective coverage of trauma care. The study used a mixed-integer nonlinear programming (MINLP) model to simultaneously locate trauma centers and helicopters while accounting for helicopter availability uncertainty. Additionally, they introduced two heuristic algorithms to solve the model and applied it to nationwide trauma care system planning in Korea, assisting government decisions on trauma system infrastructure [17]. In 2017, Shahriari et al. developed a bi-objective optimization model to determine the best locations for ground ambulance bases, aerial bases, and helipads, aiming to minimize patient transfer time while maximizing EMS service coverage. A mixed-integer programming (MIP) model was used, incorporating stochastic demand, response time constraints, and budget limitations. A case study in Lorestan, Iran, validated the model using real-world data and sensitivity analysis. The study considered three transportation cases: direct transport by ambulance, ambulance to aerial base followed by helicopter transport to the hospital, and ambulance to a helipad followed by helicopter transport to the hospital [18]. In 2018, Roislien compared optimal air ambulance base locations using population density data versus incident data in Norway. The study employed the Maximal Covering Location Problem (MCLP) model to maximize the number of demand locations covered by at least one ambulance, weighted by the number of inhabitants or the accumulated yearly incidents in each demand location. The study revealed that while the current bases are located near populated regions, the optimal bases should be positioned closer to high-incident areas [19].

There are some studies specifically focused on Ornge air ambulances. In 2013, Carnes et al. developed an optimization-based tool for Ornge to determine the optimal assignment of non-urgent inter-facility transport requests to aircraft and the optimal flight routes while considering various constraints to minimize costs. Ornge's most critical missions are unscheduled, requiring an immediate response. In 2017, a model based on MEXCLP was introduced, considering two types of aircraft (rotary-wing and fixed-wing) to optimize aircraft shift start times. Before this study, all shifts started at fixed times (7 AM and 7 PM), resulting in a short period when all aircraft were unavailable, leading to peaks in missed calls during these times.

3. Mathematical Formulation

We introduce a mathematical model that integrates the Maximum Expected Covering Location Problem (MEXCLP) with a multi-period relocation strategy to optimize aircraft locations based on demand patterns. This model accounts for relocation costs, ensuring that aircraft are repositioned efficiently while maintaining optimal coverage for both on-scene and interfacility demands. The relocation cost for air ambulances is influenced only by the distance between origin and destination bases and the aircraft speed. In this study, we assume all vehicles have the same speed, so the relocation cost can be translated into relocation distance.

If a demand point is within a predefined distance from a base location, then that demand point is covered by that base location.

We use a multi-period relocation approach, meaning we optimize the location of ambulances by relocating them at the beginning of each time period. In this model, we consider two types of missions: interfacility and on-scene, as well as two types of vehicles: fixed-wing aircraft and rotor-wing aircraft. Interfacility missions can be handled by both rotor-wing and fixed-wing aircraft, while on-scene demands can only be handled by rotor-wing aircraft. Therefore, the model aims to maximize coverage for interfacility demand using both types of aircraft and maximize on-scene coverage with rotor-wing aircraft, while penalizing the relocation distance.

Sets and parameters:

I	: Set of potential base locations, $I = \{1, 2, \dots, m\}$.
J	: Set of all demand points, $J = \{1, 2, \dots, n\}$.
T	: Set of time periods, $T = \{1, 2, \dots, h\}$.
A	: Set of possible RW coverage levels $A = \{0, 1, 2, \dots, p_1\}$.
B	: Set of possible FW coverage levels $A = \{0, 1, 2, \dots, p_2\}$.
p_1	: Number of RW (rotor-wing) aircraft in the system.
p_2	: Number of FW (fixed-wing) aircraft in the system.
q_{tRW}	: Busy fraction for RW aircraft during time period t .
q_{tFW}	: Busy fraction for FW aircraft during time period t .
D_{jtOS}	: Demand weight for on-scene calls at demand point $j \in J$ during time period t .
D_{jtIF}	: Demand weight for interfacility calls at demand point $j \in J$ during time period t .
$d_{ii'}$	: Distance between base $i \in I$ and base $i' \in I$.
d_{ij}	: Distance between base i and demand point $j \in J$.
w	: Relocation cost weight.

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d_{ij}	: Distance between base i and demand point $j \in J$.
k	: Maximum distance for coverage between base i and demand point j .
$coverset_j$	: Set of bases are within the coverage threshold for demand point j .

The decision variables of the mathematical model are given below.

Decision variables:

y_{jabt}	= Binary variable indicating whether demand point j is covered by exactly a RW and b FW aircraft during time period t .
x_{itRW}	= Number of RW aircraft stationed at base i during time period t .
x_{itFW}	= Number of FW aircraft stationed at base i during time period t .
r'_{iitRW}	= The number of RW aircraft relocated from base i to base i' during time period t .
r'_{iitFW}	= The number of FW aircraft relocated from base i to base i' during time period t .

The mathematical formulation of the problem is as follows.

Mathematical model:

$$\text{Max} \quad Z = \sum_{t \in T} \left(\sum_{i \in J} (D_{jtIF} \sum_{a \in A} \sum_{b \in B} y_{jabt} (1 - q_{tRW}^a q_{tFW}^b) \right. \quad (1)$$

$$\left. + D_{jtOS} \sum_{a \in A} \sum_{b \in B} y_{jabt} (1 - q_{tRW}^a) \right) - w \sum_{i \in I} \sum_{i' \in I} d_{ii'} (r_{i'tRW} + r_{i'tFW})$$

subject to

$$x_{itRW} = x_{i(t-1)RW} + \sum_{i' \neq i} r_{i'itRW} - \sum_{i' \neq i} r_{iitRW} \quad i \in I, t \in T \quad (2)$$

$$x_{itFW} = x_{i(t-1)FW} + \sum_{i' \neq i} r_{i'itFW} - \sum_{i' \neq i} r_{iitFW} \quad i \in I, t \in T \quad (3)$$

$$\sum_{a \in A} \sum_{b \in B} y_{jabt} = 1 \quad j \in J, t \in T \quad (4)$$

$$\text{let } C_{jtRW} = \sum_{i \in \text{coverset}_j} x_{itRW}, \quad C_{jtFW} = \sum_{i \in \text{coverset}_j} x_{itFW}, \quad j \in J, t \in T \quad (5)$$

$$C_{jtRW} - a \leq M. (1 - y_{jabt}) \quad a \in A, b \in B, j \in J, t \in T \quad (6)$$

$$a - C_{jtRW} \leq M. (1 - y_{jabt}) \quad a \in A, b \in B, j \in J, t \in T \quad (7)$$

$$C_{jtFW} - b \leq M. (1 - y_{jabt}) \quad a \in A, b \in B, j \in J, t \in T \quad (8)$$

$$b - C_{jtFW} \leq M. (1 - y_{jabt}) \quad a \in A, b \in B, j \in J, t \in T \quad (9)$$

$$x_{iFW} \geq 0, x_{iRW} \geq 0, y_{jab} \in \{0, 1\}, r_{ii'} \geq 0 \quad i \in I, j \in J, a \in A, b \in B \quad (10)$$

The objective function maximizes the expected coverage while penalizing relocation costs. The first part consists of two components: maximizing coverage by RW aircraft for on-scene demands and maximizing coverage by both RW and FW aircraft for interfacility demands. The second part of the objective function penalizes relocation distance. Constraints (2) and (3) ensure that the number of RW and FW aircraft at base i at any given time t is updated based on the number of aircraft at location i at time $t - 1$ and the aircraft that relocate to and from base i . Constraint (4) ensures that only one combination of the number of FW and RW aircraft can be assigned to each demand point at any given time t . Constraints (6), (7), (8), and (9) ensure that if y_{jabt} is selected and equals 1, then the total number of RW aircraft on the bases covering the demand point j is a and the total number of FW aircraft on those bases is b .

4. Computational Study

4.1 Dataset

In this study, we use a dataset consisting of a network of demand points and potential ambulance sites for research on optimizing ambulance deployment and redeployment [24]. All data are gathered from the Moroccan context (Fez-Meknes region). The dataset contains the locations of 199 demand points, three sets of potential base locations, the travel time between demand points and base locations, and the distances between the base locations themselves. It also includes demand data for two different time periods. Since there is no traffic limitation for air ambulances, the only parameters affecting response time are distance and aircraft speed. We consider the vehicles' speeds to be the same, so response time can be translated to distance. Therefore, we consider the travel time table as travel distance, and we assume that if a demand point is within a defined threshold distance from a base, it would be covered by that base location.

To prepare the data for our purposes, we split the demand into two types: interfacility demand, which our objective function aims to maximize coverage with both types of vehicles, and on-scene demand, for which the model maximizes coverage only with RW aircraft. In order to reduce the decision variables of our model and solve it faster, we consider only 99 of the 199 provided demand points. For base locations, we created a set that contains 6 base locations from the provided potential locations.

4.2 Implementation, Parameters

Our model is coded in Python using the Pyomo package, and we solved it using Gurobi. We set the MIPGap parameter to 0.1, enabling the solver to produce solutions more quickly by accepting solutions within 10% of optimality.

We assumed there are 6 FW aircraft and 6 RW aircraft in the system. For the initial positions, we assumed they are located at bases 1 to 6, with one FW and one RW aircraft at each base. q_1 and q_2 represent the busy fractions of aircraft, which should be calculated from historical data. The busy fraction can be estimated by dividing the total estimated duration of calls for all demand points by the total number of available ambulances. Since we did not have data regarding this, we considered a fixed value of 0.3 for both. We test different values for the relocation distance weight to determine the most effective balance, as there is a trade-off between relocation cost and maximizing coverage. We consider the coverage threshold to be 80.

4.3 Discussion of Results

Table 1 shows the location of aircraft over time. Initially, there are 2 aircraft from each FW and RW at each of the 6 base locations. Then, based on the demand of time period 1, the system optimizes the locations and relocates the aircraft considering different relocation cost weights.

Afterward, based on the new aircraft locations and the demand for time period 2, the system performs optimization again and relocates the aircraft, taking into account the different relocation cost weights. When the relocation cost weight is low, the system relocates more aircraft to maximize coverage. By increasing the relocation cost weight, the number of relocations decreases. As shown in Table 1, the number of relocations for RW and FW is higher when the relocation cost weight is low, compared to when it is high. There are no relocations when the relocation cost weight is 0.1.

Table 1: Position of Aircraft Across Bases

	initial	w = 0.0001		w = 0.01		w = 0.03		w = 0.1	
		period 1	period 2	period 1	period 2	period 1	period 2	period 1	period 2
		RW		RW		RW		RW	
Base 1	1	0	0	0	0	1	1	1	1
Base 2	1	2	2	1	1	1	0	1	1
Base 3	1	0	0	1	0	1	0	1	1
Base 4	1	3	3	2	3	2	4	1	1
Base 5	1	0	0	1	0	1	1	1	1
Base 6	1	1	1	1	2	0	0	1	1
		FW		FW		FW		FW	
Base 1	1	0	0	0	0	1	0	1	1
Base 2	1	1	0	1	0	1	1	1	1
Base 3	1	3	3	3	4	1	2	1	1
Base 4	1	0	0	1	1	1	0	1	1
Base 5	1	1	1	0	0	1	1	1	1
Base 6	1	1	2	1	1	1	2	1	1

Figure 1 shows the objective function values and the two components of the objective function: the coverage component and the relocation cost, for Case 1 in time period 1. It also displays the

relocation distances. As shown in Figure 1, increasing the relocation cost weight reduces the objective function value, as the model can no longer easily relocate aircraft with a low penalty to maximize expected coverage. Increasing the relocation cost weight does not significantly change the penalty, with the variation range being about 4, but the relocation distance decreases considerably, from about 4000 to 0. This leads to a decrease in the objective function value, particularly in the coverage component of the objective.

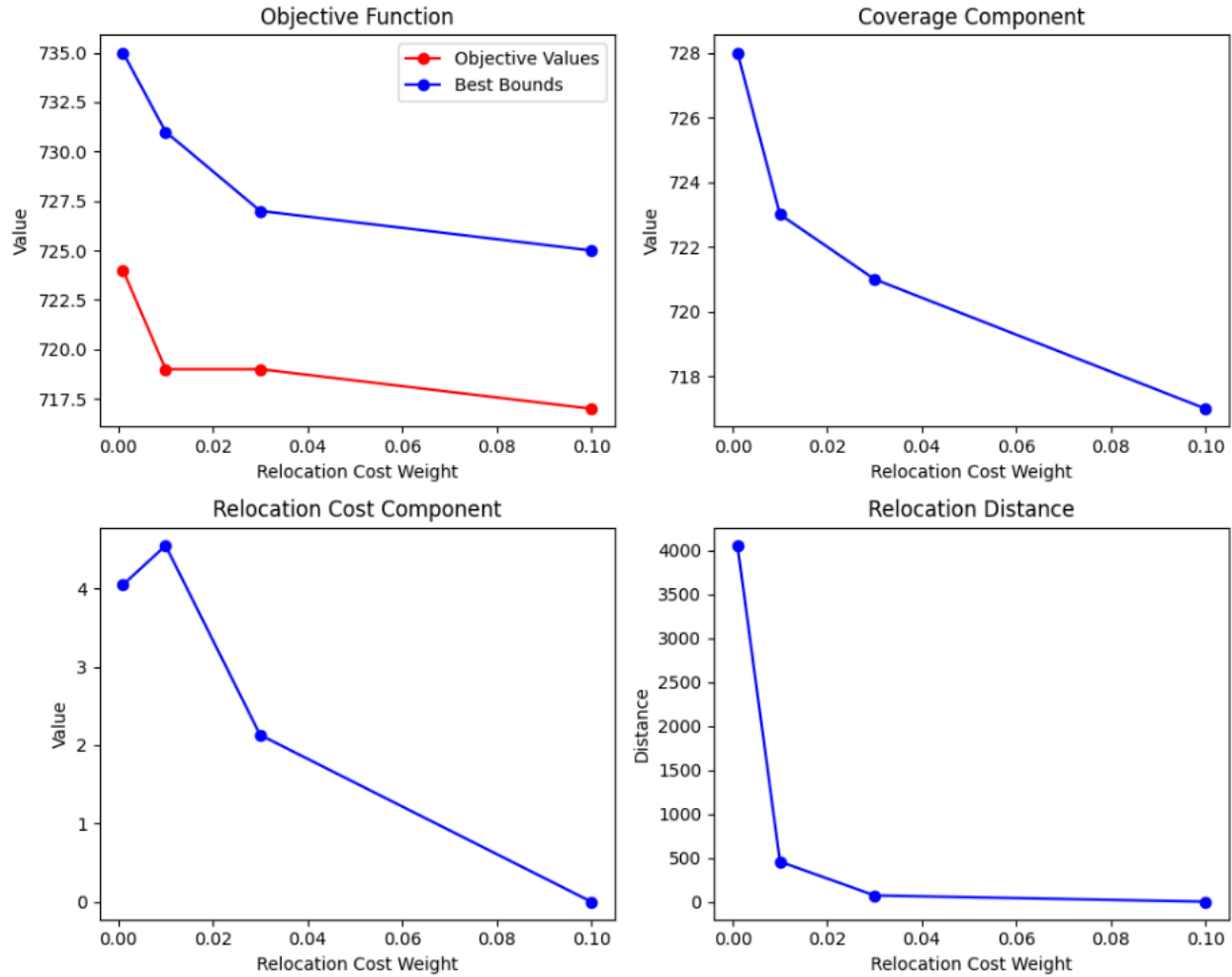


Figure 1: Objective function value for time period 1

Figure 1 shows the objective function values and the two components of the objective function: the coverage component and the relocation cost, in time period 1. It also displays the relocation distances. As shown in Figure 1, when the system performs relocations, both the objective function and coverage increase, indicating that relocation can enhance KPIs such as coverage and availability. Additionally, increasing the relocation cost weight reduces the objective function value, as the model can no longer easily relocate aircraft with a low penalty to maximize expected coverage.

One important observation is that when the relocation cost weight is 0.001 (which is very low), the relocation distance is very large. This occurs because the MIPGap is set to 0.1 and not 0, and the result is good enough not optimal. Since the relocation cost is low, the system relocates a large number of aircraft. In Table 2, you can see a comparison of using MIPGap values of 0.1 and 0 for two different values of w in time period 1. When w is 0.001, with MIPGap set to 0.1, the objective function is close to the optimal solution, but the number of relocations is very high because it doesn't impact the objective function significantly. However, when w is 0.1, since the relocation cost is high, the model minimizes the number of relocations. The results for MIPGap values of 0 and 0.1 are very similar in this case.

Table 2: Objectives using different MIP gap values for period 1

		MIPGap	
	w	0.1	0
Objective Value	0.001	724	730
Relocation Cost		4.058	730.8
Coverage		728	0.7
Number of Relocations		21	4
Objective value	0.01	719	727
Relocation Cost		4.55	3.16
Coverage		723	730
Number of Relocations		3	3

Figure 2 shows the objective function value and its component values in time period 1, similar to Figure 1. As in period 1, in period 2, the objective function value and coverage gradually increase as the relocation cost weight decreases. The demand pattern in the second period is very similar to that in period 1, so fewer relocations are needed. However, due to the gap between the best bounds and the objective function, which indicates the accuracy of the model, relocations still occur. In the first and last points, where the best bound and objective function are close, very few relocations are made.

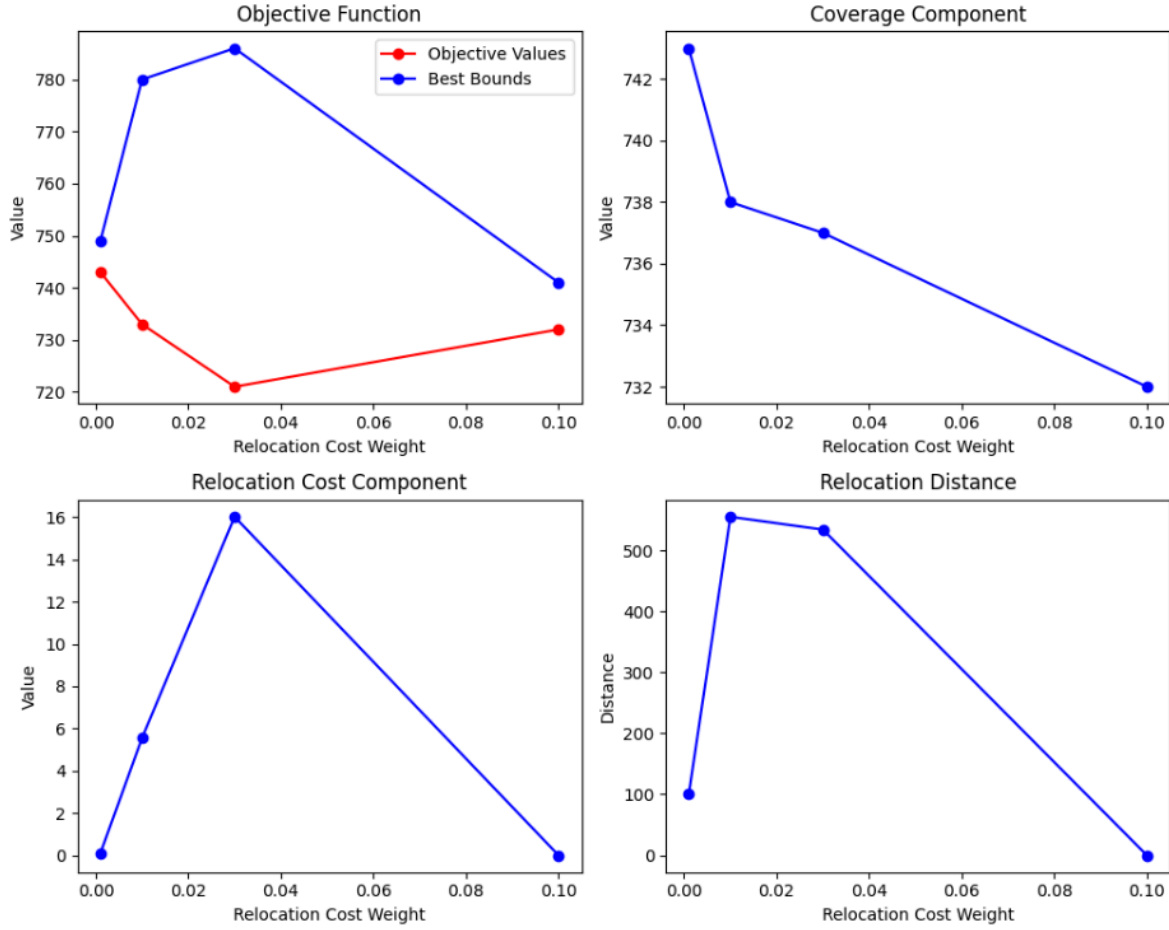


Figure 2: Objective function value for time period 2

In Figure 3, the coverage is shown before and after optimization for the first demand period with a relocation cost weight of 0.001. As shown, in the initial state, aircraft are stationed at all bases. However, after relocation, the system reallocates aircraft to bases that cover more demand, which increases the objective function, as seen in Figure 1. The reason for this is that the demand covered by a base that becomes empty can be covered by other bases. By relocating aircraft to a base that can cover these demands as well as additional demand, the overall coverage increases, which is what occurs in this case.

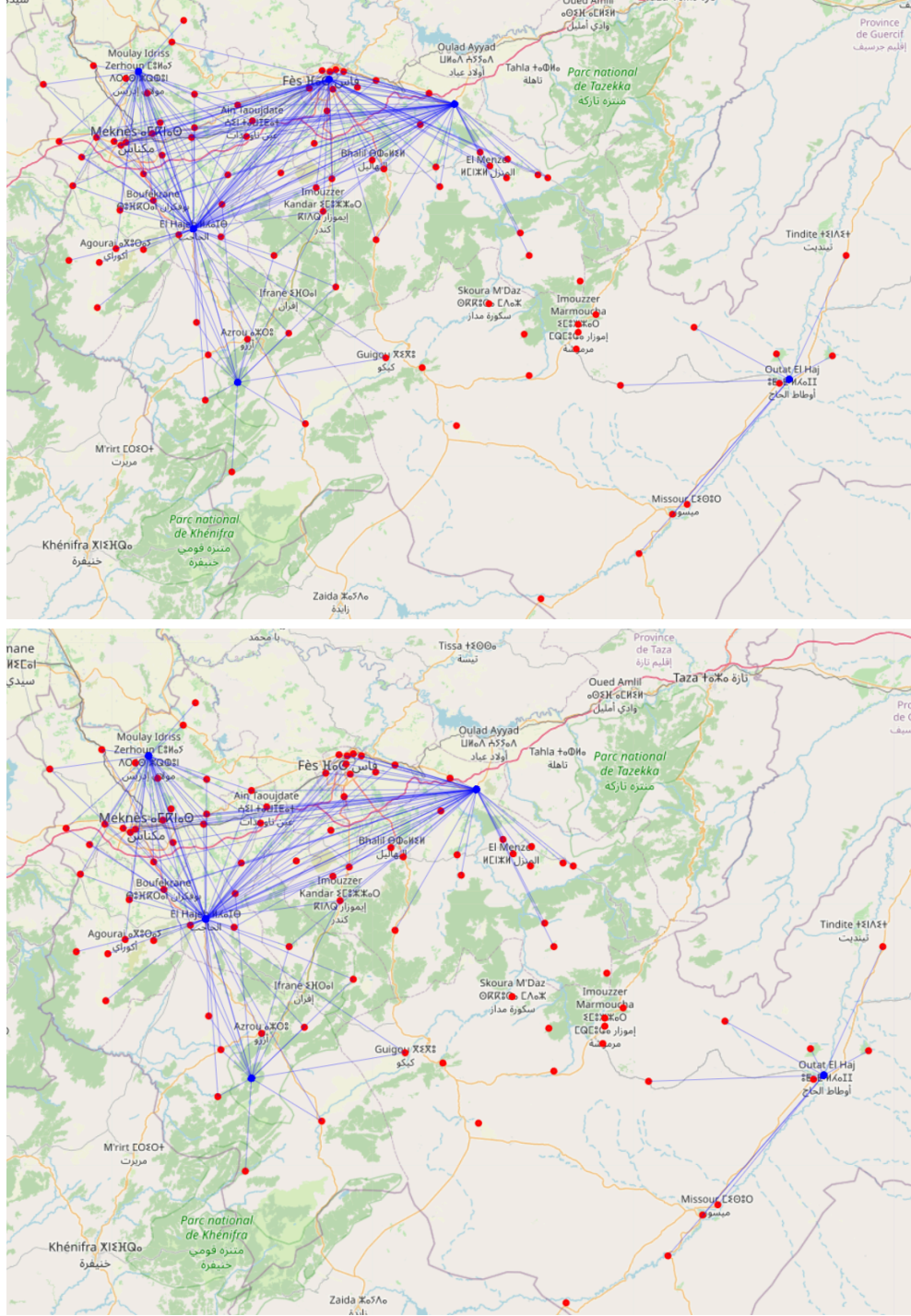


Figure 3: Coverage before and after optimization for Period 1 with $w = 0.001$.

5. Conclusion

This research introduces a mathematical optimization model for the multi-period relocation of air ambulances, designed to maximize coverage while minimizing relocation costs. The model accounts for two types of vehicles and two types of demand: interfacility demand, which can be

handled by both fixed-wing and rotor-wing aircraft, and on-scene demand, which can only be addressed by rotor-wing aircraft. By incorporating a relocation penalty, the model balances operational cost-effectiveness with the need to improve response times for unscheduled medical missions.

The results, based on real-world Moroccan air ambulance data, demonstrate that allowing aircraft relocations significantly enhances coverage. Efficient positioning of air ambulances can lead to faster response times and better resource utilization, ultimately improving patient outcomes.

For future research, we plan to focus on optimizing the location and relocation strategies for Ornge air ambulances. This study will consider two types of missions with different levels of urgency, ensuring that the model prioritize the high-priority missions. Additionally, we aim to develop a dynamic relocation model that optimizes aircraft positioning in real-time based on operational events, such as dispatching an ambulance to a call or completing a patient transfer to a hospital. This approach will allow the system to continuously adapt to changing demands, improving the overall effectiveness of emergency medical services.

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